SAMPLE

Great Observatories: Herschel Space Observatory

The Far-Infrared Explorer

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Free sample: Introduction & Chapter 1

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Introduction

Some of the universe's most important work happens in the cold and the dark. The places where stars are born, the dust clouds where planets condense, the dim glow of galaxies when the universe was young: all of it radiates not in visible light but in the far infrared, wavelengths that the Earth's atmosphere blocks almost completely from the ground. For most of the twentieth century that territory was a rumor, glimpsed by balloon and aircraft flights and by telescopes that could see only slices of it. The mission this book describes threw the window wide open: the Herschel Space Observatory, a European observatory that carried, for its four working years, the largest single mirror ever put in space and the coldest instrument chamber ever flown for astronomy.

Named for the musician and astronomer who discovered infrared radiation itself, Herschel was launched in 2009 and operated until 2013, stationed far from Earth at a gravitational balance point, chilled by a bath of liquid helium a few degrees above absolute zero. From that cold vantage it mapped the sky in light no human eye can see: the warmth of star-forming clouds, the glow of interstellar dust, the fingerprints of water and other molecules across the galaxy and far beyond it. Its datasets are still being mined years after its death, and no successor of its size has yet flown, which gives this book a slightly melancholy subject: an observatory whose like has not been seen again.

The story has everything a good spacecraft story should have. There is the engineering: a mirror too large for any existing rocket fairing until its builders bent conventional materials to their will, a cooling system that turned the whole telescope into a cryogenic instrument, and a three-instrument payload that divided the far infrared among a camera, an imager, and the most sensitive radio receiver ever put at such wavelengths. There is the science: the first views of star formation as a process that builds filaments and beads across whole clouds, the discovery that comets carry ocean-like water, the detection of water vapor around a dwarf planet, and the census of dust-lit galaxies in the universe's first chapters. And there is the ending, a planned death, as the helium boiled away and the observatory went warm, its last hours spent on farewell calibrations aimed at objects chosen with a care that says much about the people who fly such machines.

This volume follows the series pattern: the discovery and history of the mission, the understanding of what it was and how it worked, the science beneath its instruments, the observation and measurement of its results, its important features, its context in the space observatory era, its modern

legacy, the research it enabled, its significant discoveries, its challenges and open mysteries, its future in both the archive and the succession plans, its trivia, and finally its significance. The chapters are written to be evergreen, describing the mission's permanent facts and its settled science, so that the book will read as well in twenty years as today.

A note on scale before the history begins, since it helps every later chapter. The observatory weighed over three metric tons at launch, stood about seven and a half meters tall, and carried its great mirror at the focus of a telescope whose sunshield spread like a sail behind it; it observed wavelengths between about fifty-five and six hundred and seventy-two microns, four to ten times longer than the limit of human sight in wavelength, and it did so from a station so far from Earth that its signals took a twentieth of a minute to arrive home. Those numbers are the machine's skeleton; the chapters that follow put flesh on them.

A reading note, in the series' tradition. The chapters that follow keep the technical account in plain words, reserving the machinery for the two chapters that earn it, and they keep the story chronological where history leads and thematic where the science does. Names appear only where they are permanent: the mission's institutions and its founders' family, the instruments' acronyms spelled out once, and the classes of objects the observatory made famous. Everything volatile, the mission's current citation counts, the shifting politics of its succession, is either stated as of its era or left to the reader's own updating, so that the book's account is one that does not go stale on the shelf.

One more note on the book's evidence, in the series' tradition of stating sources plainly. The account that follows rests on the mission's own publications: the instrument description papers, the survey definition papers, the operations records, and the mission's obituary literature, published in the professional journals in the years since the helium's end, plus the archive's own documentation, which is public and permanent. The reader who wants the figures behind a given sentence will find them in those places, cited by the mission's own name and its papers' years, and the book has kept to facts that exist in that record, which is the discipline that lets a volume like this promise to be evergreen: it repeats only what the record settled.

Chapter 1: The Discovery of Herschel Space Observatory

The discovery of Herschel as a mission was a decades-long negotiation with the laws of physics, and the story begins with the man whose name it carries.

William Herschel was a Hanoverian musician who emigrated to England, taught himself optics, and built telescopes good enough to find the planet

Uranus in 1781 from a back garden in Bath. Nearly two decades later, in 1800, he performed the experiment this mission honors: passing sunlight through a prism, he measured the temperature across the rainbow of colors and found the thermometer climbing beyond the red edge, in light no eye could see. He called it calorific rays; we call it infrared, and the discovery founded the science this telescope served. His sister Caroline, his collaborator and an astronomer of independent distinction, completes the family portrait, and the mission that took the family name eventually repaid the sister's memory too, in a gesture saved for its final days and told later in this book.

The far infrared, the portion of the spectrum Herschel opened that runs from roughly fifty microns to a fraction of a millimeter, is where the universe hides its coldest business. Star-forming clouds are around thirty degrees above absolute zero, and their warmth peaks in the far infrared. Dust, the raw material of planets, glows there. Water and many other molecules have their strongest lines there. But the Earth's atmosphere is a black curtain across the band, absorbed by water vapor and carbon dioxide, and worse, a telescope warm enough to live at room temperature blinds itself, glowing exactly at the wavelengths the astronomer wants to see. Two conditions follow: the telescope must go above the atmosphere, and it must be made colder than the objects it studies. Both are hard, and the second is very hard.

The decades of preparation tell the story of the band being opened by inches. Ground-based telescopes on high dry mountains caught the shortest slivers. Airborne observatories put telescopes on aircraft and, in the 1970s and 1980s, flew the first serious far-infrared instruments above most of the water vapor. Balloons carried small cryogenic telescopes to the edge of space, and a pair of small satellites, one Dutch and American, one European, made the first orbital surveys of the infrared sky, teaching the trade at low resolution. By the 1980s the scientists of the European Space Agency and its member institutes were sketching the obvious next step, an orbiting observatory with a genuinely large, genuinely cold telescope, dedicated to the band nothing else could open properly.

The project's road was long by the standards of any era. The mission was first proposed in the early 1980s, adopted as a cornerstone mission of the agency's science program in 1993, and spent the rest of the decade and the start of the next in redesign, as its planned arrangement with a French-led infrared mission collapsed, its budget was trimmed, and its design was folded into a shared flight. The final configuration paired it with the Planck microwave observatory on a single rocket, a double launch of two major space telescopes that was itself a first, and the pair left Earth together in May 2009 from the coastal range in South America. The spacecraft's working name during much

of its development honored the astronomer who first mapped the infrared moon, and it was renamed for the Herschel family before flight, a change that made the name say exactly what the telescope did.

The discovery, in the end, was that the far infrared could be opened the way visible light had been, with a flagship-class observatory of long lifetime and open competition for its time. That had been in doubt. The band's previous steps had been small, specialized, and brief; Herschel was general-purpose, large, and international, and its success demonstrated what the community had argued for across three decades: that the cold universe deserves a flagship telescope of its own. The argument has an uncompleted sequel, discussed in the later chapters on the future, but the discovery itself stands: the window opened, and a family of experiments begun in a Bath garden in 1800 finally received its great observatory.

The launch itself deserves its closing paragraph in this history. The double payload left the South American coastal spaceport in May 2009 on a heavy rocket, and the two telescopes separated from the upper stage in turn within the first hour, the infrared observatory released first and its microwave partner minutes later, after which both set out for their common distant harbor. The first months were a commissioning marveled at in every account of the mission: the cryostat performed, the instruments woke cold, the first test images showed star-forming regions in light no machine had ever seen properly, and the announcement that the observatory was open came with pictures that made the invisible sky famous overnight. The mission had been thirty years in the argument, and its first images settled the argument's final doubts in an afternoon.

A paragraph on the mission's name history, small but persistent in the literature. In its long design years the observatory was known by the name of a fourth-century astronomer, the last great astronomer of antiquity, whose star catalog and solar theory closed the classical tradition; the infrared mission that name had belonged to was cancelled in its early years, and the name passed through the project's early documents like an heirloom. The

renaming for the Herschel family, decided in the years before launch, gave the telescope the only name it could rationally have carried, and the old name survives only in the histories, where its mention signals a document's age to anyone who knows the sequence.

A last note on the history's shape, for the reader who will meet other missions' stories in this series. Herschel's thirty-year road from proposal to launch was long even by the standards of its era, and the honest history records that the road's length was mostly the price of the machine's ambition: the mirror, the

cryostat, and the instruments were each pushed past the state of the art, and the schedule stretched while the state of the art caught up. The pattern, of a mission arriving late because it is arriving large, is common in the great observatories, and the book keeps the point here so the later chapters' superlatives carry their provenance: nothing about this machine was early, and everything about it was once impossible.

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